

Life-cycle assessment of denim

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4.1 Introduction

Sustainable fashion started before the Industrial Revolution, when people conserved resources and the cost and labour it took to produce the basic things necessary for life [1]. The raw material for textile fabrics came from nature (fibers and colourants) and it required a long time to transform these into garments. At that time, the requirements and cost of labour were huge; hence it was affordable only for the wealthy. By the end of the 18th century, industrial machines were introduced to spin yarns and weave fabrics, which accelerated the production process of textiles. In later years, synthetic dyes, synthetic fibres and other new products of the Industrial Revolution allowed the textile industry to grow quickly. Around the mid 19th century, textile products became affordable for all classes of people, and apparel was more abundant and easily accessible, which led to the development of consumerism. Therefore, the purchasing capacity of consumers increased the rate of production of textiles and garments. With the growth in the textile and apparel industries came many factors, including consumer awareness, government intervention by legislation, and the diversification of textile production. Since 1980, owing to the development of information technology, consumers have been able to access data showing the impacts of this industry. Thus, the attention of consumers has forced manufacturers and retailers to take action regarding the environmental, economic and social impacts of various processes of the textile value chain [2,3].

During the Industrial Revolution, textile manufacturers were neither aware nor concerned about the environmental effects of the production of textiles, and they disposed of chemicals into nearby rivers and watersheds. Subsequently, the emergence of environmentalism and social consciousness took place. Consumers began to consider new ways of disposing, recycling and reusing their clothes. They became concerned about how their clothing was produced and the impact this process had on the environment, and thus developed a conscientious lifestyle. As a response to conscientious consumerism, eco-labeling arose as a way to inform consumers that companies cared and were on board with a sustainable viewpoint. Over the past decade, the global society has begun to focus more on environmental issues (i.e., global warming) and its impact on all fields. Therefore, manufacturers have had to develop sustainable products rather than disposable ones. This chapter focuses on the life-cycle assessment (LCA) of denim jeans, because we are aware that denim is a significant part of the fashion industry as a result of its popularity. According to the report by Cotton, Inc., US consumers typically own seven pairs of denim at a time [4].

The LCA is a detailed analysis of a product; it describes the entire life span of products, from fibres to the retailing of denim. Therefore, it encompasses the production of raw materials, the manufacturing process of denim garments, logistics and the garment's use, and ends with the denim's reuse, recycling and disposal. The product's use phase and end-of-life disposal phase depend on consumer behaviour; nevertheless they are critical stages in a product's life cycle. During the manufacturing phase of denim, at every stage there is an impact on the environment, such as energy and water use, CO₂ emission and waste [5,6].

4.1.1 Stages of a life-cycle assessment

Generally the LCA can be divided into four stages. The first stage includes defining the goals of the study and describing the product (denim) to be assessed. The second stage is inventory; data from all processes of the product's life cycle are collected and processed, from the acquisition of raw materials to the development of the final product. During this stage, energy, raw material requirements, environmental emissions and discharges relating to the product can be computed and presented. These data can be used to calculate the discharge from all of the processes of the product's life cycle.

The third stage is called the 'impact assessment stage', in which the inventory data are translated into their effects on human health, ecological health and resource depletion. The last stage of this process may be called 'recommendations' based on an analysis of the impact assessment results. These are the basic stages of LCA assessment, but they may vary for different raw materials, modified production processes and the choice of one product over another [6–9]. An LCA is an important procedure which helps us assess the environmental impact of any product. Hence, we can easily identify the most polluting stages of the process and use them as a guide to develop enhanced methods and processes. An LCA is a burgeoning topic and is studied by many researchers [10–12].

4.2 Product life cycle

The main purpose of this assessment is to reduce resource use and emissions into the environment as well as improve performance throughout its life cycle. Fig. 4.1 is an illustration of all phases in a product's life cycle [13].

4.2.1 Raw material procurement

Denim has always been made from cotton [14]. Sometimes synthetic materials are used to produce denim to provide additional properties such as comfortable fit and durability. We know that cotton is one of the most popular natural fibres used in the world. Three percent of the world's cultivated land is used for cotton production. Based on previous studies on the environmental, natural fibres are good (environmentally friendly) and synthetic fibres are bad (harmful). However, each fibre is different, with its own challenges to sustainability in the production process that cannot be

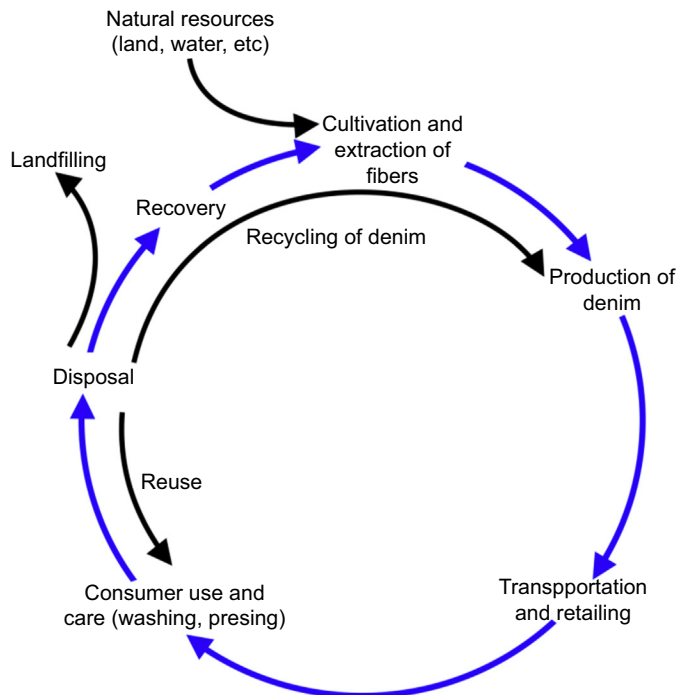


Figure 4.1 Phases of a product's life cycle (for denim).

ignored [13]. Generally, cotton is a natural fibre which is grown and not manufactured. Although cotton is an environmentally friendly fibre, it consumes huge quantities of water and requires pesticides and fertilizers to grow well for the expected quality and quantity [15]. Based on a data sheet published by the US Department of Agriculture and the Organic Consumers Foundation, cotton is the most toxic crop in the world. Cotton uses more than 25% of all insecticides in the world and 12% of all pesticides [16,17]. The majority of these insecticides are considered the most toxic chemicals in the world. The health risks of pesticide exposure include birth defects, reproductive disorders and weaker immune systems [18–22]. Also, cotton requires 7–29 tons of water to produce 1 kg of raw cotton [23–25]. As a result of these environmental and health issues, most denim brands are committed to organic cotton materials [26,27]. As to synthetic materials, polyester and polyurethane are the most used in denim manufacturing. Although these fibres require less water and land to manufacture, they have effects on the environment. Generally, synthetic polymers are obtained from nonrenewable resources (e.g., fossil fuels) and the production of these fibres requires an enormous amount of energy, causing higher greenhouse gas (GHG) emissions. Environmental problems in waste management such as nonbiodegradability pose various health and toxicological threats that also have negative impacts created during the production of synthetic fibres [7].

4.2.2 Manufacturing

Manufacturing is the process in which fibres are converted into yarn and then fabric, and finally are converted into garments (three-dimensional structures). Fig. 4.2 shows

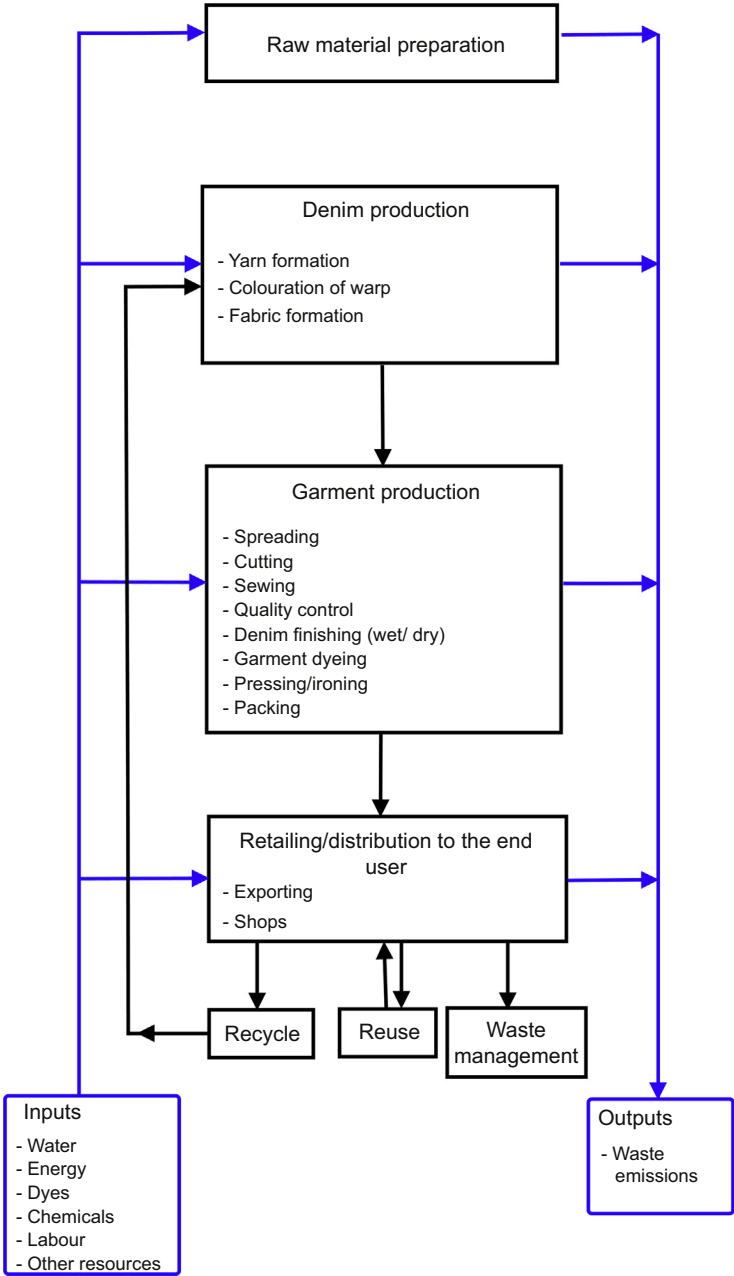


Figure 4.2 Production life cycle of denim.

a map of the each process related to denim manufacturing with respect to the inputs and outputs of the production chain for the product life cycles.

4.2.3 *Spinning*

Yarn formation for denim fabrics is carried out by two methods: ring spinning or open-end spinning. Over the past 3 decades, open-end yarns have dominated in the field of denim fabric production. Generally, cotton and other fibres are produced or manufactured in different places and/or countries, and the first phase of yarn formation is to transport fibre from the place of harvest to the spinning factory. Several processes are carried out (i.e., opening, carding, combing, drawing, roving, spinning and winding) to form yarn, which require a high level of energy use. Dust, fibre and yarn waste production are main issues related to environmental protection during spinning [7,28]. Based on the condition of the fibre, there is a possibility for the use of lubricants, chemicals, water, packaging materials and other disposables during the various spinning processes, including *blowing*, *carding*, *drawing*, *combing*, *roving*, *ring spinning* and *open-end spinning*. The highest environmental and health risk in this process is the inhalation of the dust, which may cause a dangerous disease called byssinosis (commonly called ‘brown lung’) [7,29].

4.2.4 *Fabric formation*

Denim fabric is obtained by weaving twill structures. After the warping process (beaming), it is necessary for the warp sheets to undergo a sizing process in which a sizing box contains natural or synthetic sizing agents such as polyvinyl alcohol or carboxymethyl cellulose. The fabric manufacturing process requires high levels of energy consumption and produces waste from various chemicals, water and packaging materials. The disposal of this waste, as well as excessive noise, can cause issues [30–33].

4.2.5 *Colouration of denim*

Colouration is a process that is accompanied by major environmental impacts owing to the production of dyes, the use of water and chemicals and the fixation of the dyes and the drying of denim as well as nondenim fabrics. These textile materials need pretreatment before colouration. During pretreatment, they are cleaned with sodium hydroxide and detergents to remove natural hydrophobic substances. Washing of these process lines may cause water pollution [34]. For denim fabrics, only the warp yarns are dyed. Generally, indigo and sulphur dyes are used to colour warp sheets of denim. After dyeing, the fabric is rinsed with water, excess dye is removed and dried (water pollution results from coloured wastewater) and then sized. In the dyeing bath, the indigo dye has to be chemically reduced using sodium hydrosulphite as a reduction agent. The reduced indigo (with a yellow colour) will adhere to the fibre. The reduced dye will oxidize back to the characteristic blue colour when exposed to air [35–37].

4.2.6 Garmenting

The final stage of the denim supply chain is ‘garment manufacturing’, also known as the cut–make–trim (CMT) stage. During this process, various operations such as spreading, cutting, sewing, heat pressing, ironing and packaging are involved. Unlike yarn and fabric manufacturing, the CMT process involves mostly manual human labour to complete the garment. Important considerations here in the environmental impact assessment are the high level of energy use; fabric waste during the cutting process (often already recycled); accessory waste; rejected garments; lubricant, water and chemical use; use of packaging and hanging materials; and the production of high-carbon emissions during transport operations [7,13].

As mentioned earlier, the production stages from raw materials to finished textile goods are generally located in different places and countries, which make logistical activities an important component along the denim supply chain. Transportation of goods is not related only to manufacturing; once manufacturing is completed, the denim is usually sent to wholesalers, retailers, distribution centers and shops using different kinds of transportation (i.e., road, maritime or air), all of which have different amounts of energy consumption values and CO₂ emission levels. CO₂ emissions from freight transport are estimated at around 8%. In reference to the LCA, transportation to consumers is a huge factor which consumes large quantities of oil or gas, and the price of oil and gas has been steadily increasing, thus increasing transport costs. It is important to reduce the amount and distance of the transport of goods, popularly known as ‘green logistics’, to ensure the management of a green supply chain [38–41].

4.2.7 Consumer use and disposal

Well-meaning people who truly want to protect the environment can be passionate about what should be done with what they consider to be a generic material, called waste. The LCA process is now geared toward the consumer and is no longer solely within the manufacturer’s control (e.g., consumers can pass valued garments on to family, friends or charity or send to for reselling or recycling) and varies among and within regions and cultures. This phase encourages consumers to buy clothes only when they are needed and to take care of them to prolong their life. However, most denim clothing is washed in water every 2 weeks and then disposed of into a landfill after 1 year. Most companies put much effort into the care of denim; for example Levi Strauss & Co.’s ‘Care Tag for the Planet’ reminds consumers to wash in cold water and wash less. However, you cannot make an egg-omelet without having eggshells to dispose of. The main motto of the disposal stage, ensuring recycling and donating to reduce the amount of waste going into landfills, helps to affect the environment positively. The most commonly known waste management strategies are the 3R’s: *reduce*, *reuse* and *recycle*. The meaning of reuse is simply using the product for the same purpose or for something completely different. Reusing includes donating unwanted items to charity groups or even to a family member or friend. The main benefits of reducing and reusing are to prevent pollution caused by developing new materials (i.e., producing or cultivating new fibres such as cotton or polyester). Also, they save energy and money, reduce GHG emission and global climatic change, and most important,

help sustain the environment for future generations. The meaning of recycle is to collect and process materials that are considered trash and turn them into new products. Recycling saves resources and uses less energy than producing new items. Levi Strauss & Co. started the '8 Bottles. 1 Jean' concept, recycling eight polyethylene terephthalate bottles (typically beverage bottles) to produce one pair of jeans [42]. A few companies extracted the fibre from old denim (considered trash) for technical applications such as insulation for homes and civic buildings.

4.3 Life-cycle assessment

An LCA is an analytical tool for the systematic and quantitative evaluation of the environmental impacts of a product or service system throughout all stages of its life. It is an undeniable fact that denim products require the use of raw materials, energy, equipment and labour. Generally, these key points determine design and product development. Thus an LCA is necessary for an analysis of details about product design by assessing the environmental and economic impacts during the product's life cycle, with an aim of ensuring sustainable development. An LCA studies environmental aspects and potential impacts throughout a product's life (i.e., cradle to grave) from raw material acquisition through production, use and disposal. LCAs are critical for understanding the global progress of reducing negative impacts on the environment. Currently, an LCA is the most comprehensive approach to assessing the environmental impacts of textile products and is graded by the International Organization for Standardization (ISO) 14040-14043 industry standard [43–46]. An LCA does not include social or economic impacts of sustainability. The LCA framework has described as having four phases, shown in Fig. 4.3. This assessment is based on the definition of its goal and scope, inventory analysis, impact assessment and interpretation of results [13,47,48].

4.3.1 Goal and scope

The goal and scope are the first phase of an LCA, which sets the context of the study. In them, the following items should be considered and clearly described:

- the functions of the product system or, in the case of comparative studies, product systems
- the functional unit
- the product system to be studied
- the system boundaries
- allocation procedures
- types of impact and the methodology of impact assessment, and subsequent interpretation to be used
- data requirements
- assumptions
- limitations
- initial data quality requirements
- type of critical review, if any
- type and format of the report required for the study

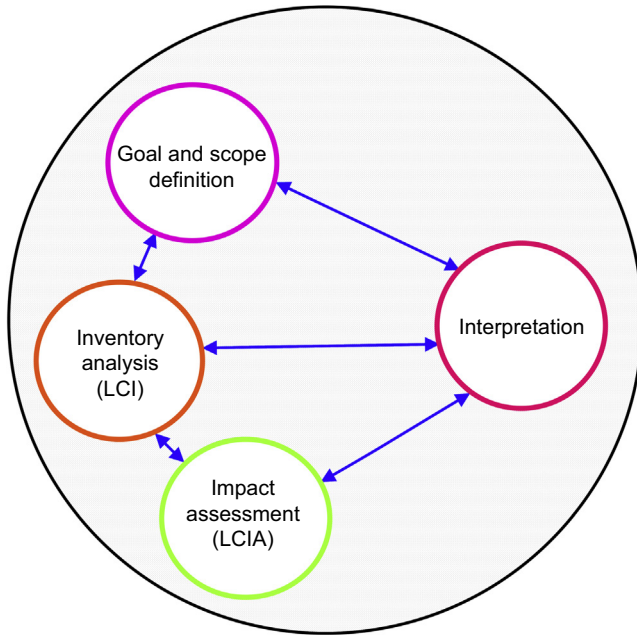


Figure 4.3 The life-cycle assessment framework [47,49].

Adapted from sources – Ali S, Life cycle assessment, submitt. as coursework phys. 240, Stanford Univ. Fall 2010; 2010. <http://large.stanford.edu/courses/2010/ph240/ali2/>; Rebitzer G, Ekvall T, Frischknecht R, Hunkeler D, Norris G, Rydberg T, Schmidt W-P, Suh S, Weidema BP, Pennington DW, Life cycle assessment: part 1: framework, goal and scope definition, inventory analysis, and applications, Environ Int 2004; 30:701–20. doi:10.1016/j.envint.2003.11.005.

The LCA is an iterative technique; therefore, the scope of the study may require modification according to the product design and other parameters of the study being conducted as additional information is collected.

4.3.2 Life-cycle inventory analysis

A life-cycle inventory (LCI) analysis is the second phase of an LCA; it is the data collection portion. It is straightforward accounting and involves the ‘system’ of interest, consisting of the detailed tracking of all process flows for the entire system (i.e., from raw materials to final products, energy use, water, carbon, and other substance emissions) based on ISO 14040:2006. Therefore, it is a complex analysis which may involve dozens of individual unit processes in the supply chain (e.g., extraction of cotton or other fibres, yarn formation, colouration, fabric formation, garmenting, finishing, transportation, retailing and distribution) as well as hundreds of tracked substances.

4.3.3 Life-cycle impact assessment

The impact assessment is done after the inventory analysis. In general, this phase can focus on the results of the inputs and outputs and strategically categorize them,

helping to analyse the environmental impact. Various methods are available globally to categorize and characterize the life-cycle impact of the flows to and from the environment, but it can complicate the comparability of different LCA studies. Apart from this, there are other variables in LCI assessment (LCIA) including the system boundary, the functional unit and specific LCIA methods such as allocation. Mandatory elements for the LCIA are:

- the selection of impact categories, category indicators, and characterization models
- the classification stage, in which inventory parameters are sorted and assigned
- impact measurement, indicating where the inventory in categories is characterized, using many possible LCA methodologies

4.3.4 Life-cycle interpretation

This is the last phase of LCA, in which we ‘identify, quantify, check and evaluate’ information from the results of the previous phases such as LCI and or the LCIA [50]. This study aids in ensuring the validity of the entire study. ISO 14040:2006 requires the interpretation phase to:

- identify significant issues based on the results of the LCI and LCIA phases of an LCA
- evaluate the study, considering completeness, sensitivity and consistency checks
- have conclusions, limitations and recommendations [13,50].

4.4 Life-cycle assessment of denim

In 2004, Levi Strauss & Co. began a green campaign involving sustainable fashion. They initiated many case studies with respect to their product. In 2007, they conducted an LCA analysis to assess the environmental impact of a pair of Levi’s jeans, which are made from cotton. The analysis was made from cradle to grave (seed sowing to landfill). The study opened up a new era for awareness regarding the environmental issues caused by one pair of jeans. The environmental impact was assessed in the following categories:

1. cotton production
2. fabric production
3. garment manufacturing
4. transportation and distribution
5. consumer use
6. recycling, which then goes back to Step 2, or
7. waste stream in a landfill

From this work, an idea was born to create sustainable fashion in their supply chain (design, production, merchandise and marketing), which ensured the reduction of environmental impacts. Therefore, the primary objective of the LCA approach was to provide designers and developers with the information they needed to produce more sustainable products. A secondary objective was to provide a scientific method to support claims of the environmental improvement of products. Although not an initial objective of this effort, it was later realized that

the methodology also provided a rigorous means to communicate environmental performance to suppliers (Levi Strauss & Co.) [51]. The LCA study on denim conducted by Levi Strauss & Co. was based on the ISO 14040 standards, in which all significant processes from ‘cradle to gate’ was analysed based on one pair of Levi’s 501 jeans. The phases are:

- fibre cultivation (sowing to harvesting)
- yarn formation and fabric formation (ginning, spinning, preparation for weaving and weaving)
- colouration (colouration of warp)
- cut, sew and finish (garmenting and garment washing)
- transport and retail (transportation at each stage and distribution to retail stores)

Table 4.1 shows the impact categories that were investigated to carry out an LCA.

Commercially huge number of software is available to measure LCA, but Levi’s used ReCiPe 2008 for this study [53]. This assessment helped Levi’s and other brands in the apparel sector to use more effective and holistic approaches to address the environmental impact caused by this industry.

Several data were included in cycle mentioned earlier. For example, in the spinning phase, the type of fibre, origin of the fibre, and fibre loss during processing were taken into account. In the case of colouration and weaving, the type of dye (indigo, sulphur, pigment, etc.), weave structure, and sizing efficiency were considered. During the cutting and sewing phase, the cutting efficiency (i.e., marker efficiency), materials, sundry material/weight, and packaging material/weight were included. The types of chemicals used during the finishing phase were considered; finally, the transport mode and distance were considered and collected from all phases of the entire supply chain.

Table 4.1 Impact categories and descriptions [52]

Environmental impact categories		
Category	Description	Units
Climate change	Global warming potential of greenhouse gases released into environment	CO ₂ emissions (kg)
Water consumption	Net freshwater taken from environment minus water returned to the same watershed at the same quality or better	Litres
Eutrophication potential	Oxygen depletion as a result of nitrogen and phosphorous deposit into freshwater or marine environments	PO ₄ emission (g)
Land occupation	Total land occupied to support product system assessed	m ² /year
Abiotic depletion	Measure of depletion of nonrenewable sources including fossil energy, metals and minerals	Sb emissions (mg)

Levi Strauss & CO. The life cycle of a jean; 2015. p. 16–50. <http://levistrauss.com/wp-content/uploads/2015/03/Full-LCA-Results-Deck-FINAL.pdf>.

4.4.1 Functional unit

The functional units are the number of items to be analysed for this study. Therefore the functional unit for this study is the production of one pair of Levi's 501 medium stonewash jeans (women's) with a weight of 340 g. Details of the functional unit are listed in [Table 4.2](#).

4.4.2 System boundaries

The boundaries of denim jeans systems from the cradle-to-gate phase of the life cycle are: raw materials extraction, fabric formation, colouration, garment manufacturing, transportation, retailing and distribution.

4.4.3 Material phase

The material phase covers how much cotton cultivation is required to produce one pair of denims. We know that cotton cultivation requires a huge quantity of fertilizers, water and pesticides to prevent insects and worms, and herbicides to prevent weeds. This study phase includes procedures used to extract cotton (harvesting, transportation and ginning) to produce one pair of denim jeans.

4.4.4 Production phase

The production phase consists of more process lines which convert the raw materials (fibre) into a finished product (denim jeans). The number of processes is directly proportional to the higher amount of environmental impact. In this phase, different processes are involved in making final products, i.e., yarn formation, fabric formation, colouration, garment manufacturing and the finishing process.

Table 4.2 Functional unit of life-cycle assessment (LCA)

Functional unit	Pair of Levi's 501 women's jeans (one pair)
Material	100% cotton
Raw material	Cotton made in United States, Brazil, India, China, Pakistan and Australia
Dyeing	Indigo blue
Finishing	Medium stonewash
Weight of jeans	340 g
Year of production	2012
LCA conducted	2013
LCA published	2015
Consumer phase	United States, United Kingdom, France and China
Study conducted by	Industrial Ecology Consultants and Levi Strauss & Co.
International Organization for Standardization (ISO)	ISO 14040
Number of studies conducted	Not disclosed

4.4.5 Environmental impact categories

The environmental impact potential is expressed at the midpoint and end-point levels. A description of environmental impact categories is shown in [Table 4.1](#). The following categories were adapted from the LCA of a pair of Levi's 501 denim jeans conducted by Levi Strauss & Co., based on the ReCiPe 2008 methodology. The essential environmental impact categories are: ozone depletion, terrestrial acidification, human toxicity, photochemical oxidant formation, particulate matter formation, terrestrial eco-toxicity, freshwater eco-toxicity, marine eco-toxicity, ionizing radiation and natural land transformation [\[52\]](#).

4.4.6 Climate change

Climate change is one of the major concerns of the 21st century. Because it is one of the major polluting industries of the world, the textile industry has drawn intense attention to its affects on climate change, which results in global warming, a rise in sea levels and extreme weather. Increasing consumption of textiles for clothing is causing the biggest impact on the environment. In the case of denim, from fibre production to finished products, meaning throughout the supply chain, the textile industry emits a huge amount of carbon, initiating a change in the climate. These changes usually affect the ecosystem as well as human health, and they increase every year. This study focuses on the global warming potential of radioactive forcing, and measuring the effect of emissions of CO₂ and other GHGs into the environment. The final result can be expressed in terms of how it affects human health and the ecosystem. Some facts about cotton cultivation are that:

- The production of 1 kg of cotton (equivalent to a pair of jeans) can take more than 20,000L of water.
- As much as 1.5% of the world's annual energy is consumed by industrial fertilizer production. Because it is a key consumer of fertilizer, cotton has a huge role in global climate change.

4.4.7 Water consumption

The World Bank estimates that almost 20% of all freshwater pollution is made by textile production, in particular wet processing. We are aware that the global textile (including denim) industry releases pollutants continuously, causing unimaginable harm to the environment. Generally, it pollutes the land and makes it useless. A survey showed that cotton crops consume the highest amount of harmful pesticides and fertilizers, which pollute and spoil the water and land. Thus water consumption is an important factor affecting human health and the ecosystem. The life cycle for LCA is expressed at the midpoint level simply because no model can provide the end-point level. As per the ReCiPe 2008 LCA tool, water consumption can be calculated from raw material procurement to the end product. Therefore, this study focused mostly on water that is lost and damaged rather than

recycled. There are some facts which highly correlate cotton cultivation and water consumption:

- The Aral Sea, which is the world's fourth largest lake, has been reduced to 15% of its original size as a result of irrigation for the cotton industry.
- A total of 73% of the global cotton harvest comes from irrigated land.

4.4.8 Eutrophication

Eutrophication is an environmental problem that affects water bodies. Eutrophication occurs as the result of a process in which high nutrient levels stimulate the growth of aquatic plants, especially algae, on the surface of still or slow-moving water bodies. The growth of algae on the water surface reduces the penetration of sunlight, reducing oxygen levels. Also, as the result of photosynthesis, algae reduce oxygen levels, so both processes affect the water bodies. The resulting level of deoxygenation is increased by microbial action upon dead algae on the floor of a lake or estuary. Ultimately these limitations in the available levels of sunlight and oxygen reduce the diversity of plant and fish life that the affected waters can sustain. In LCA, eutrophication is expressed at the end-point level for damage to the ecosystem. Generally phosphorus is an essential nutrient for plants; however elevated amounts of phosphorus cause plants to soak up more nitrogen before all phosphorus is depleted, causing excessive algae growth in bodies of freshwater [13].

4.4.9 Land occupation

This study assessed how much damage to the ecosystem is caused by land occupation. In general, it concentrates on the area of land used for raw materials, cultivation and production, and the transformation of a certain area of land. Denim production requires larger areas of land than do other apparel products, owing to cotton cultivation. Therefore, in this study results are expressed at the midpoint and end-point levels, because both systems are affected [13].

4.4.10 Abiotic depletion

Abiotic depletion is the loss of nonliving natural resources such as naturally occurring minerals, fossil fuel energy, and metals. These resources, such as fossil fuels and minerals, are essential to our everyday lives and most are being extracted at an unsustainable rate, which seems to be leading to a shortage of resources for future generations. The results for denim regarding the overall costs of resources lost during cultivation and production are calculated, expressed at the midpoint and end-point levels.

4.4.11 Inventory analysis

LCA data for denim jeans was collected from Levi Strauss & Co. The results express the environmental impacts of a pair of Levi's 501 jeans explained in [Section 4.4](#). These



Figure 4.4 Life cycle of a pair Levi's 501 jeans [36].

Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

analyses were made with the help of the ReCiPe 2008 tool. Based on this tool, the input and output inventory was calculated with respect to environmental impacts. Fig. 4.4 illustrates the life-cycle phase of denim from cradle to grave [13,52].

4.5 Life-cycle assessment results

4.5.1 Climate change

Generally, climate change results from the emission of GHG and is followed by impacts on global warming. The emission of GHGs in textile production is substantial, resulting in an increase in the temperature of the system and damage to human health, animals and entire ecosystems. To produce one pair of denim jeans, consumer care was considered to have the highest impact on global warming; it measured 37% on the life cycle. Fabric production had the second largest impact on global warming (27%). The third biggest impact on global warming was transport and retail; its impact was 11% on the life cycle. Climate changes (expressed in kilograms and percentage) to the environment with respect to the different phases of denim production are shown in Fig. 4.5. Consumer care contributes the largest impact on climate change. CO₂ emissions are increased by processing factors such as washing with warm water, using a dryer, washing weekly and using a conventional washing machine. Results are shown in Figs 4.6 and 4.7.

4.5.1.1 Water consumption

Water is important because it is needed for life to exist; it is an abundant resource on earth. Water is used for many purposes include agricultural, industrial, household, recreational and environmental activities. Most important, water is essential for sustaining the ecosystem. Climate change will have significant impacts on water resources

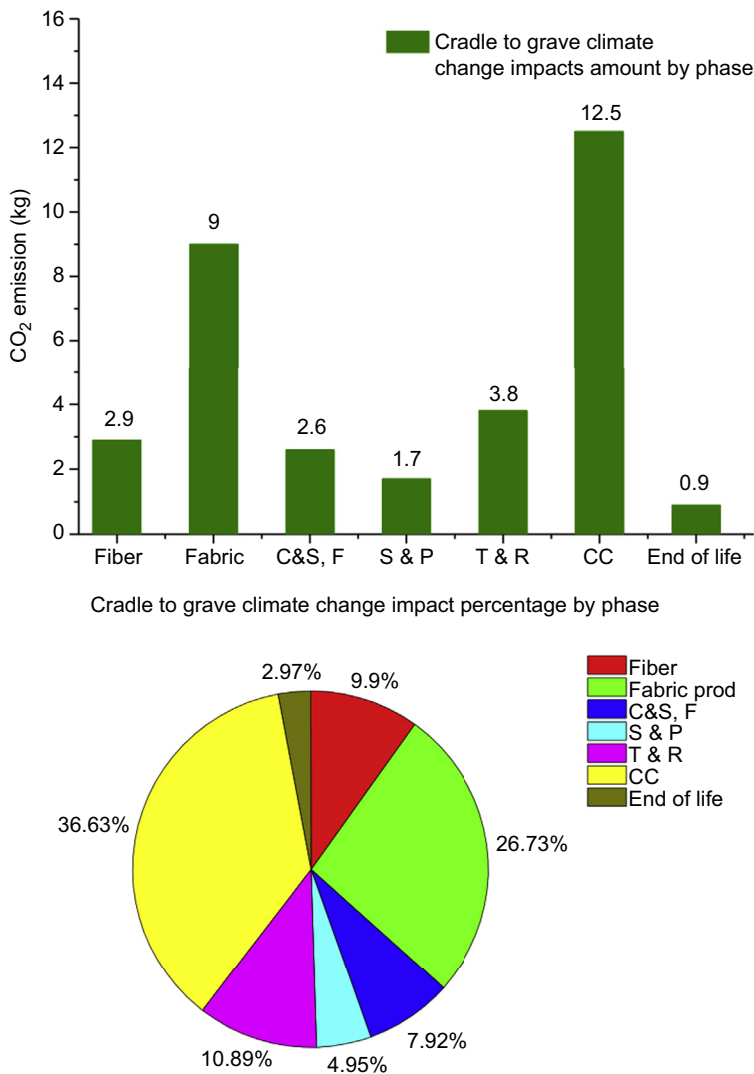


Figure 4.5 Climate changes to the environment with respect to different phases of denim production. *C & S, F*, cut and sew, finishing; *CC*, consumer care; *Fabric Prod*, yarn formation, fabric formation, colouration; *Fiber*, fibre cultivation to harvesting; *S & P*, sundries and packing; *T & R*, transport logistics and retail [52].

Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

around the world because of the close connections between the climate and the hydro-logic cycle. Because of the expanding human population, competition for water is growing so that many of the world's major aquifers are becoming depleted. Thus water resource management has become and will remain important. Levi Strauss & Co. conducted water consumption studies in major cotton producing countries such as

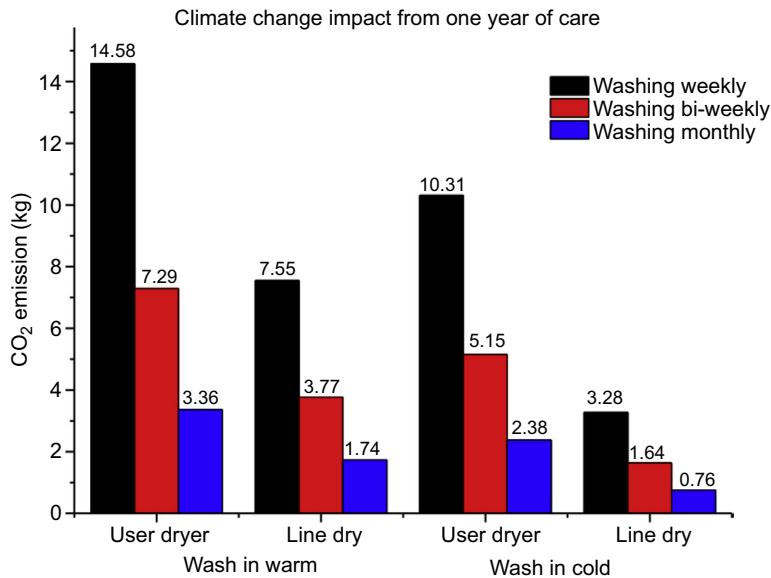


Figure 4.6 Climate change impact of 1 year of care with a conventional washing machine [52]. Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

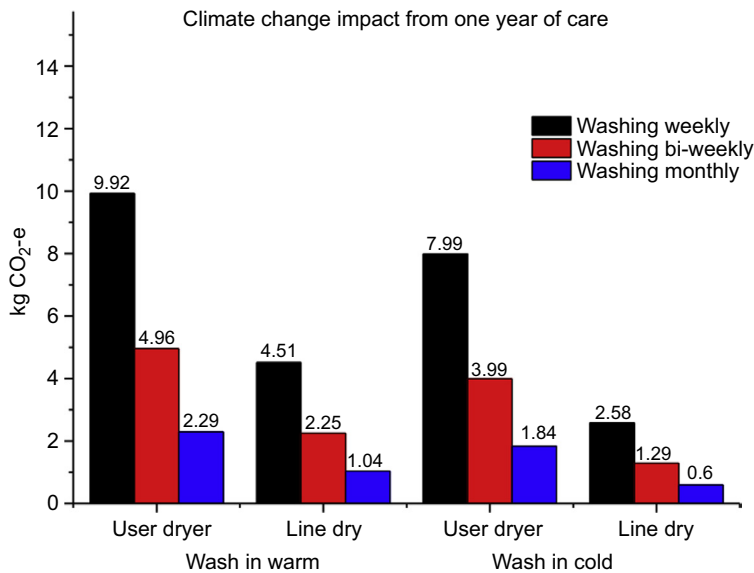


Figure 4.7 Climate change impact of 1 year of care with an efficient washing machine. [52]. Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

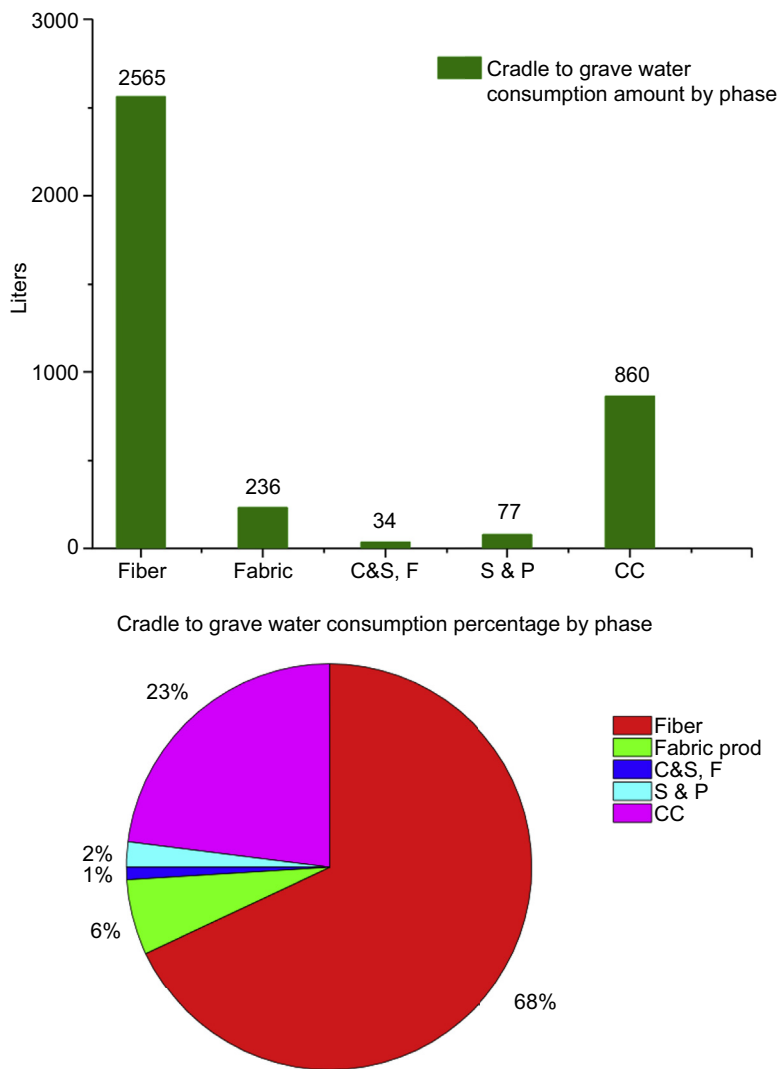


Figure 4.8 Water consumption for different phases of denim production. *C & S, F*, cut and sew, finishing; *CC*, consumer care; *Fabric Prod*, yarn formation, fabric formation, colouration; *Fiber*, fibre cultivation to harvesting; *S & P*, sundries and packing; *T & R*, transport logistics and retail [52]. Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

the United States, India, China, Brazil, Australia and Pakistan. Cotton crops consume a large quantity of water from sowing to harvesting. The study proved that fibre production consumes 68% of the entire LCA. The second largest consumption of water is during consumer care activities such as cleaning and washing, at 23% (Fig. 4.8).

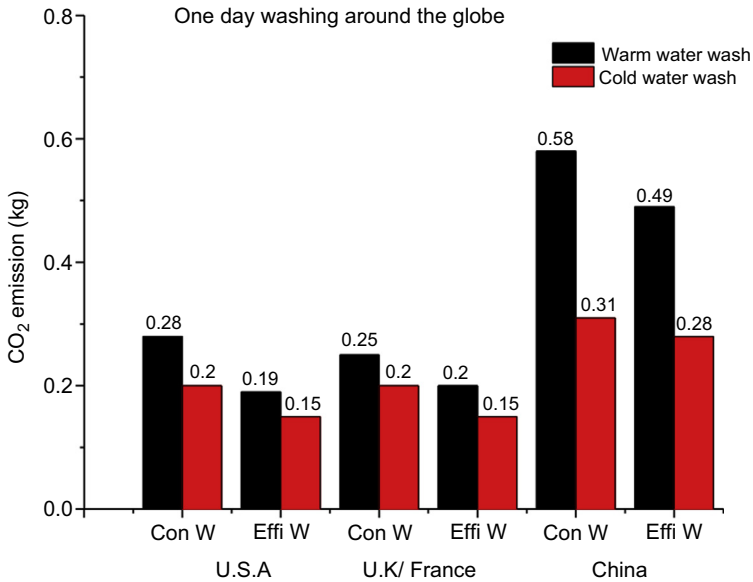


Figure 4.9 Impact of emission resulting from 1 day of washing, in different countries. *U.K.*, United Kingdom; *U.S.A.*, United States [52].

Adapted from *The life cycle of a jean*, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

Fabric production consumes 6% during the processes involved in converting fibre into yarn and yarn into fabric.

Concerning consumer care, Levi Strauss & Co. conducted various studies. From their studies, we get some idea about how consumer care can have an impact on climate change and water consumption. A ‘One Day of Washing Around the Globe’ survey was conducted to obtain more information about denim washing (consumer care phase) techniques. The emission results for the United States and the United Kingdom/France were almost same in both cases (i.e., conventional and efficient washers). However, to use warm water, we need to use more energy, which creates higher CO₂ emissions (in kilograms) than does washing in cold water. The emission results for China were two times higher than those of the United States, and it had a huge impact on environmental effects with respect to climatic changes. Overall, efficient washing techniques may cause less emission potential than conventional washing techniques. The detailed study is described in Fig. 4.9.

Levi Strauss & Co. studied water intake for 1 year’s care of denim (Fig. 4.10). Results showed that washing denim once a week consumes a large quantity of water; around 958 L per year for one pair of denim. That is two to four times higher consumption than other techniques such as washing biweekly or monthly. The most important fact from that study is that efficient washing machines consume 30% less water during washing than conventional washing machines with respect to weekly washing.

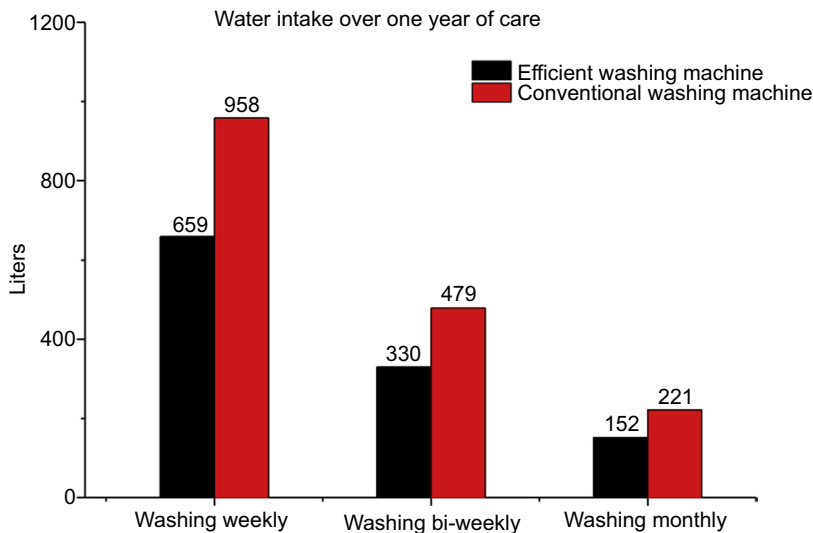


Figure 4.10 Water consumption of denim washing during consumer care, using different washing techniques [52].

Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

4.5.1.2 Eutrophication

Over the past century humanity has transformed its water bodies from a clear-water ecosystem into a eutrophic (nutrient-rich) environment. Today it is highly contaminated with nutrients that cause eutrophication, such as algal blooms, red tides, dead zones and a range of related serious problems for the ecosystem and the people living nearby and earning a living from it. It has a significant impact on the aquatic ecosystem, damaging drinking water sources, fisheries and recreational bodies of water. There is no doubt that it damages human health and the entire ecosystem. Eutrophication has been happening silently all over the world and it is a big environmental issue. In the case of denim production, the fibre cultivation stage has a huge impact on the environment, and a 37% share. The second largest eutrophication impact on the environment is consumer care, with a share of 16%. Apart from this, sundries and packing, fabric production and transportation have high impacts to the environment due to the eutrophication process. The cradle to grave eutrophication process of denim, by phase (PO_4 emission in grams and percent share by various processes) can be seen in Fig. 4.11.

4.5.1.3 Land occupation

Humans are the major force of change on the earth, transforming the land for agriculture that provides food, shelter, production systems and infrastructure for development and such. However land transformation can affect the planet's physical, chemical and biological systems and directly affect ecosystems. Therefore land is the basis for most biological and human activities on earth. The transformation rate has drastically increased

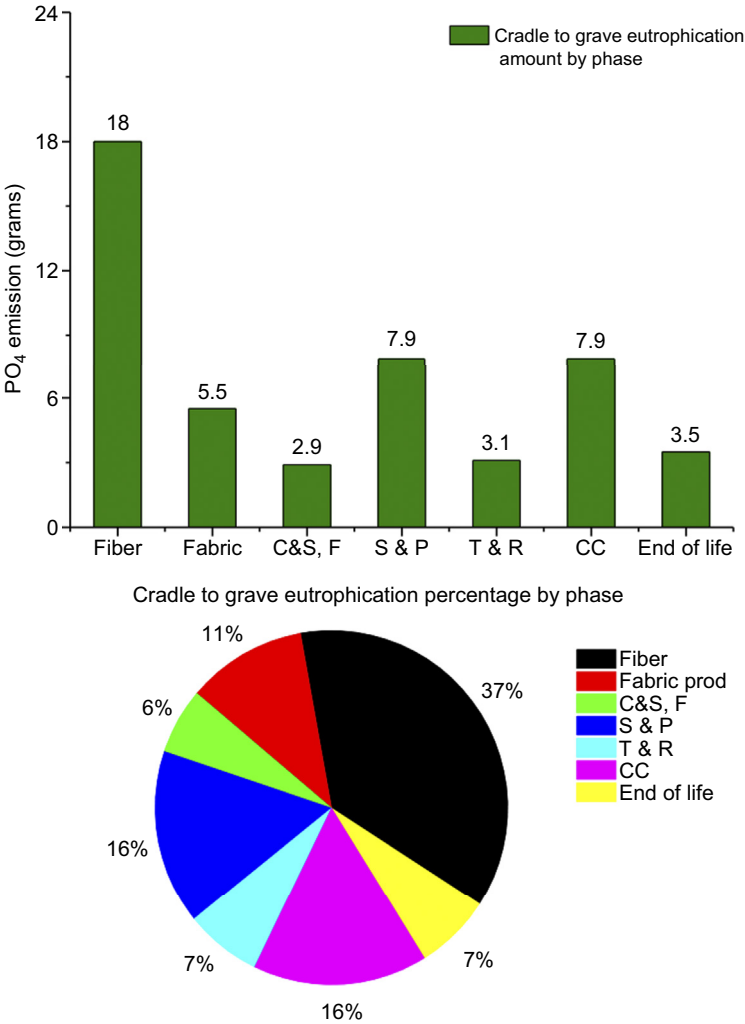


Figure 4.11 Cradle-to-grave eutrophication amount of denim by phase (PO₄ emissions in grams and percent share by various processes). *C & S, F*, cut and sew, finishing; *CC*, consumer care; *Fabric Prod*, yarn formation, fabric formation, colouration; *Fiber*, fibre cultivation to harvesting; *S & P*, sundries and packing; *T & R*, transport logistics and retail [52]. Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

since the 19th century and has had an impact on the ecosystem, global warming, floods, etc. Owing to the production and agricultural sector, land occupation is unavoidable. Compared to other fibres, cotton consumes a larger quantity of the land for cultivation to production. Its cultivation requires large areas of land as well as production activities because of the number of processes in yarn and fabric formation. To produce one pair jeans, 9.3m²/year area for cultivation purposes is required, which is 78% of the share.

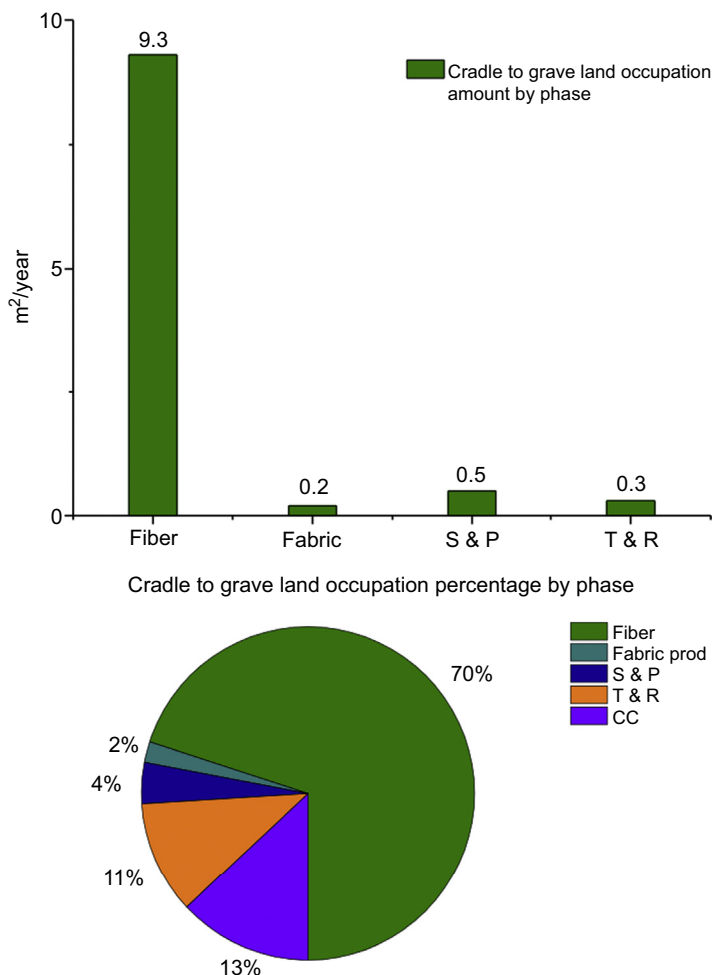


Figure 4.12 Cradle to grave estimate of land occupation with respect to the production of one pair of denim jeans. *C & S, F*, cut and sew, finishing; *CC*, consumer care; *Fabric Prod*, yarn formation, fabric formation, colouration; *Fiber*, fibre cultivation to harvesting; *S & P*, sundries and packing; *T & R*, transport logistics and retail [52].

Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

Fibre production has consumed the highest land occupation and entails consumer care, transportation, logistics and retail. The brief results are shown in Fig. 4.12.

4.5.1.4 Abiotic depletion

These are the measures of the gradual loss of natural resources such as minerals, fossil fuels and metals. Natural resources are resources that exist without the actions of humankind. However, natural resources contain renewable and nonrenewable resources.

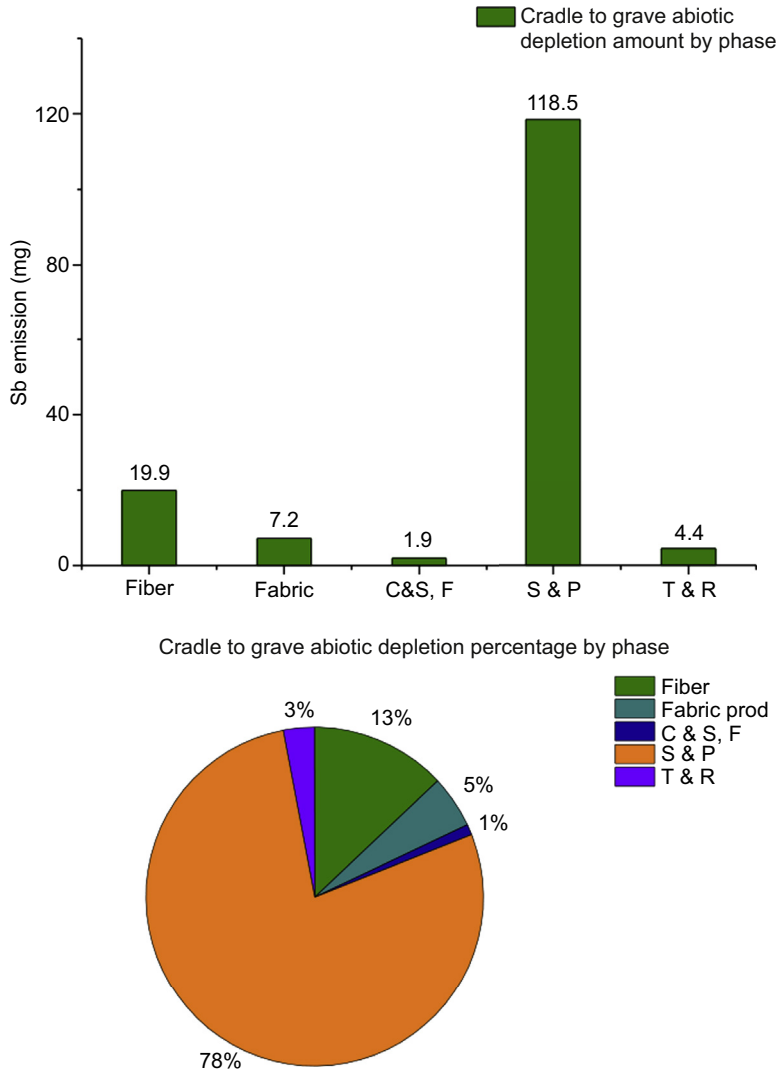


Figure 4.13 Abiotic depletion cradle to gate to produce one pair of denim jeans. *C & S, F*, cut and sew, finishing; *CC*, consumer care; *Fabric Prod*, yarn formation, fabric formation, colouration; *Fiber*, fibre cultivation to harvesting; *S & P*, sundries and packing; *T & R*, transport logistics and retail [52]. Adapted from The life cycle of a jean, by Levi Strauss & Co., 2015; Copyright 2015 by Levi Strauss & Co.

The result of abiotic depletion with respect to producing one pair of denim jeans is reported in Fig. 4.13. Over the entire life cycle, the sundries and packaging phase of the production process is 118.5 mg and accounts for 78% of the cradle to gate portion of the denim life cycle. Other phases require less; of that, fibre production

produces 19.9 mg Sb emission. Next, fabric production, cutting, sewing, finishing and transportation has a small impact on abiotic depletion.

4.6 Interpretation and conclusion of results

Climate change: In the production of one pair jeans, the fabric production and consumer care stages have the highest impact on climate change owing to the number of processes which use resources.

Water consumption: This result mainly reflects litres of water used and not reused, and that fibre production and consumer care require large quantities of water; we need a solution for this.

Eutrophication: As to eutrophication in the producing one pair of denim jeans, fibre cultivation has the highest impact. This is because of the use of fertilizer and pesticides, which damage water bodies, causing nutrient-rich runoff water. They generally affect the oxygen content in water, altering aquatic systems.

Land occupation: Results shows that the highest land occupation required to produce denim jeans is for fibre cultivation, which obviously requires a huge area. Apart from cultivation, land occupation may occur in the forest to cut trees to produce energy. This practice is still carried out in many developing countries.

Abiotic depletion: It is considered to be one of the largest environmental issues caused by textile production. This is the result of the use of natural resources to convert fibre to garments. During denim production, sundries consumed the highest resources with the greatest impact on abiotic depletion.

Materials, production and other findings:

- Fabric assembly, which includes yarn spinning, dyeing, weaving, and fabric finishing, had notable contributions related to their impact on climate change and nonrenewable energy consumption.
- Life-cycle stages that had a minimal contribution to impact include fabric transport, product transport, packaging, production waste, distribution, retail, and end-of-life waste.

The entire life cycle of one pair of Levi's 501 denim jeans is equal to:

- climate change: 33.4 kg CO₂ emissions; this is equal to a US car driven 111 km, or 246 h watching TV on a plasma big screen
- water consumption: 3781 L, or 3 days of a household's total water needs in the United States
- eutrophication: 48.9 g PO₄ emissions; this is around the total amount of phosphorus found in 1700 tomatoes
- land occupation: 12 m²/year of land, equal to seven people standing in a relaxed manner

Consumers will be given information about the environmental impact of denim production by labeling, which is another potential output of a technique such as LCA. Labeling such as eco-labels can identify better environmental policies used in products. Customer do not have much knowledge about relevant aspects of denim production and how it has been optimized. It is not necessary for them to be flooded with figures and scientific details. They need a clear message that allows them to choose products

according to the information given. In practice there are no such customs followed in eco-labeling, which is confusing and leads to misunderstanding. Eco-labeling needs to be based on established methods such as LCAs and linked to best available technology criteria and sustainability standards such as ISO 260000.

4.7 Future trends

In the future, the same methodology can be used to find LCAs for other denim garments which may be made from lyocell and other sustainable fibres [54]. It would provide ideas which would make it possible to compare the environmental impacts of similar products with respect to the different fibres. LCA research, which is available with statistical data, is needed yearly to account for changes in fibre production, a company's sustainable initiative, the availability of suppliers and factors that contribute harm to human health, the ecosystem, diversity, and resource availability.

4.7.1 *Software for life-cycle assessment*

A lot of software is available in the marketplace. The most widely circulated LCA software includes:

- Boustead Consulting Database and Software
- ECO-it: Eco-Indicator Tool for environmentally friendly design: PRé Consultants
- ECOPRO: Sinum Corporate Environmental Management
- EDIP: Environmental design of industrial products: Danish Environmental Protection Agency
- EIOLCA: Economic Input-Output LCA at Carnegie Mellon University
- GaBi 3 – (Ganzheitliche Bilanzierung): University of Stuttgart (IKP)/PE Product Engineering
- IDEMAT: Delft University Clean Technology Institute Interduct Environmental Product Development
- KCL-ECO 3.0: KCL LCA software
- LCAiT: CIT EkoLogik (Chalmers Industriteknik)
- LCNetBase: LCA using traceable US data: Sylvatica
- SimaPro 5.0 for Windows – PRé Consultants
- SPOLD: Society for the Promotion of Life-Cycle Assessment Development
- TEAM (Tools for Environmental Analysis and Management): Ecobalance, Inc.
- Umberto: Advanced software tool for Life Cycle Assessment: Institut für Umweltinformatik

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